

07-mastra-ai

07. Mastra AI Framework

Level: □ Intermediate -> □ Advanced

Jam: 12 (5 minggu x 2 sesi)

Prasyarat: JavaScript Fundamentals, TypeScript Basics, Node.js Basics

Output: AI agents with tools, memory, RAG, workflows

Tujuan

- Paham konsep AI agents vs raw API calls
- Bikin agent pake Mastra framework
- Bikin tools (custom function buat agent)
- Memory: agent ingat percakapan
- RAG: agent jawab dari dokumen
- Multi-step workflows

Materi

Sesi	Topik	Prasyarat
1	Mastra intro, project setup, agent pertama	-
2	Tools: createTool + Zod schema	Sesi 1
3	Memory: agent ingat context	Sesi 2
4	RAG: vector search + knowledge base	Sesi 3
5	Workflows: multi-step agent pipeline	Sesi 4

Contoh Agent

```
import { Agent } from '@mastra/core/agent';
import { createTool } from '@mastra/core/tools';
import { z } from 'zod';

const weatherTool = createTool({
```

```

    id: 'get-weather',
    description: 'Get weather for a city',
    inputSchema: z.object({ location: z.string() }),
    execute: async ({ location }) => {
      return { output: `${location}: 28°C, cerah` };
    },
  });

const agent = new Agent({
  id: 'weather-agent',
  name: 'Cuaca Assistant',
  instructions: 'Kamu asisten cuaca. Pake weatherTool.',
  model: 'openai/gpt-4.1-nano',
  tools: { weatherTool },
});

const response = await agent.generate('Cuaca di Jakarta?');
console.log(response.text);

```